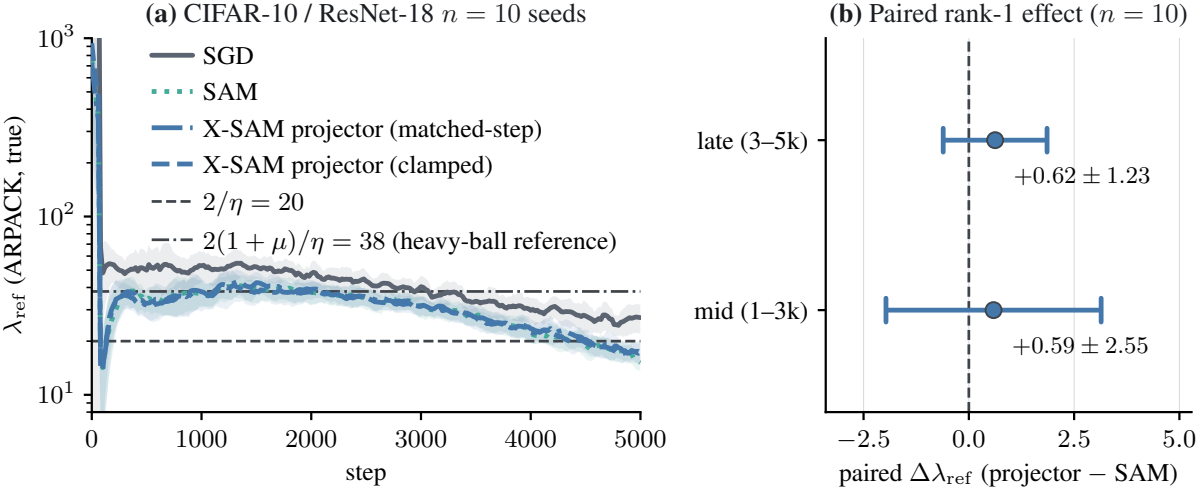

(a) CIFAR-10 / ResNet-18 $n = 10$ seeds
SGD
SAM
X-SAM projector (matched-step)
X-SAM projector (clamped)
$2/\eta = 20$
$2(1+\mu)/\eta = 38$ (heavy-ball reference)
λ_{ref} (ARPACK, true)
10^3
10^2
10^1
0
1000
2000
3000
4000
5000
step
(b) Paired rank-1 effect ($n = 10$)
late (3–5k)
mid (1–3k)
$+0.62 \pm 1.23$
$+0.59 \pm 2.55$
−2.5
0.0
2.5
5.0
paired $\Delta\lambda_{\mathrm{ref}}$ (projector − SAM)